



Eating Disorders & Neurobiology

Science and Eating Disorders

Eating disorders are serious mental health conditions that can lead to significant health problems. Thanks to new research, we now understand more about how eating disorders affect the brain, how they develop, and how we can better support people as they recover.

People with eating disorders, and the people who care about them, often wonder how the disorder started or may even blame themselves. Research helps us challenge harmful myths and better understand how complex eating disorders really are. We now know that there is no single cause of an eating disorder — for example, you don't need to have other mental health issues or past trauma for one to develop. Eating disorders actually often begin after a period of eating too little or not getting enough nutrition, whether this happens on purpose or because of stress or illness.

The Role of Genetics in Eating Disorders

Things like mood, personality, anxiety, self-control, appetite, body weight, and metabolism are strongly influenced by our genes (i.e. they are inherited characteristics). On average, about half of a person's risk of developing an eating disorder comes from their genetics, but this amount is different for everyone. People with a high genetic vulnerability to eating disorders may not necessarily develop an eating disorder, but are at higher risk than those without the genetic vulnerability.



The Gene-Environment Interaction

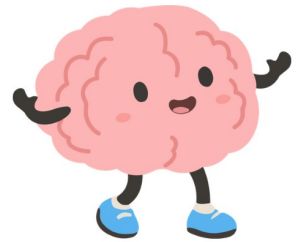
"Epigenetics" is the study of biological mechanisms that cause our underlying genetic predispositions to be "switched on" or "switched off". In certain environments, especially where there is a lot of stress or weight loss, the risk is higher — and the genes might get "switched on". As international eating disorders expert Professor Cynthia Bulik explains: "Genes load the gun, environment pulls the trigger".

Western culture places a high value on appearance and many people dieting or excessively exercise to change the way they look. For some people, these behaviours result in very few negative consequences. For others (with high genetic risk), these environmental influences can trigger their genes to "switch on", and result in an eating disorder. For example, two individuals might get ill with a stomach bug resulting in some weight loss. One person may naturally regain the lost weight with no long-term consequences, while for the other, the development

of an eating disorder may be triggered. Dieting is a known trigger to eating disorders.

How Eating Disorders Affect the Brain

Some people worry that eating disorders are caused by a chemical imbalance in the brain, but there is no evidence for this. However, research has shown that brain activity can be affected by even a small amount of dieting, and a young person's developing brain is particularly vulnerable. When a person's brain is not adequately fueled, they may struggle to make decisions, solve problems and regulate their emotions. See our handout on [Starvation Syndrome](#). They may also experience perceptual disturbances in the way they see themselves; for example, looking in the mirror and see themselves as much larger than they actually are. Also, although eating disorders aren't caused by a chemical imbalance in the brain, restricted eating and excessive weight loss can result in problematic changes to our brain chemistry. For example, the brain produces less serotonin, which results in increased symptoms of depression.



What Does All This Mean For Recovery?

The good news is that the effects of starvation can be reversed with renourishment. This can be challenging though, as increasing food intake can be scary for someone with an eating disorder.

Brain-imaging studies show that brains can adapt and change throughout recovery. We can "exercise" and change the brain by learning and practicing new ways of thinking and relating to others. Over time, people who recover from an eating disorder often show brain activity that becomes more similar to people who have never had one. A starved brain won't adapt or function optimally, so the first priority is always to improve nutrition. Also the brain tends to "lag behind" the body in terms of recovery and it can take time for people recovering from an eating disorder for their brain to "catch up" and think differently.

Unfortunately, our society can be a triggering environment, with images of unachievable bodies and inaccurate and conflicting messages about diets and exercise. Treatment helps people to understand their genetic vulnerabilities and supports them to develop skills and manage environmental influences. It is important to remember that recovery from an eating disorder *is* possible. With adequate re-nourishment and learning, the brain and body can return to healthy functioning.